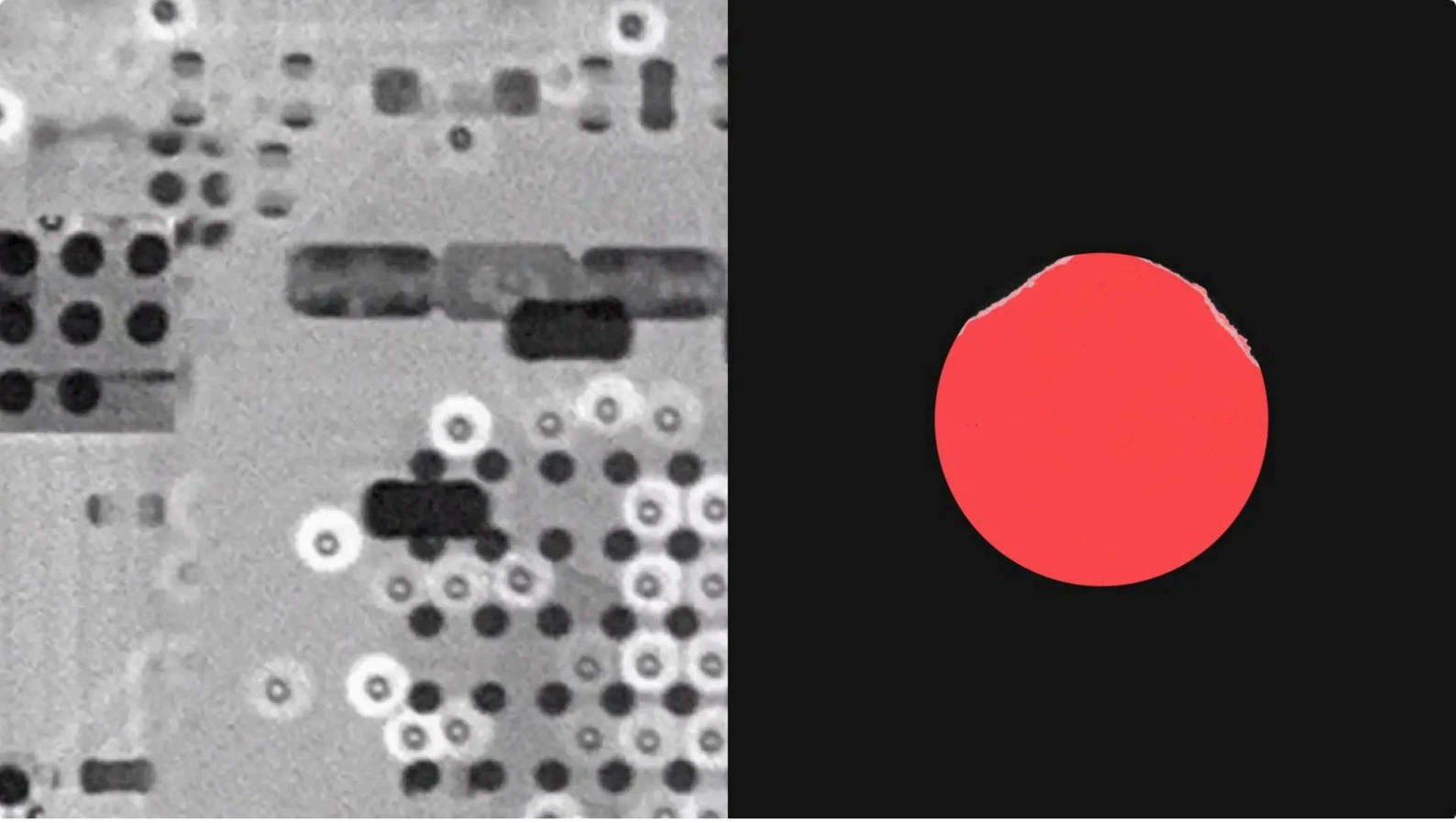


OBLIVION / TECHNICAL RESEARCH

THE GLM MODEL FAMILY

A technical review of Z.ai models, from GLM-130B to 5.3 and specialist branches

Engineering research · Architecture comparison · Critical assessment for Oblivion

26 October 2026 | Research edition 1.1

GLM is a family, not a single model

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GLM belongs to Z.ai (formerly Zhipu AI), not Anthropic. The family has expanded from pretrained language models into reasoning, agentic coding, vision, OCR, speech and image generation. Comparing these distinct branches with one generic intelligence score or parameter count is technically unsound.



Conceptual infrastructure image for the report narrative; not verified photography of a Z.ai data center.

1M	744–753B	320B / 18B	0.9B
Nominal context · GLM-5.2 / 5.3	Reported total size · GLM-5.3	Total / active · GLM-5.3-Flash	Approx. size · GLM-OCR

Engineering verdict for Oblivion

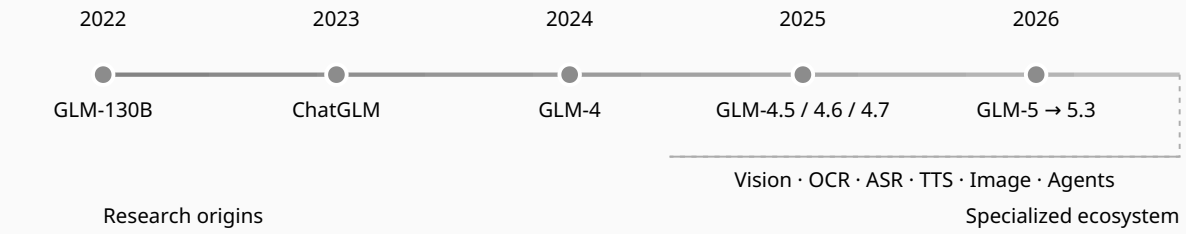
GLM-5.3 is the current text flagship for complex coding and long-horizon work. **GLM-5.3-Flash** is a new-base multimodal branch, not an 18B lightweight model. For constrained local hardware, **GLM-4.7-Flash** and small specialists such as **GLM-OCR** are more realistic starting points. Route vision, document and speech workloads to dedicated branches.

> **Final position**
>
> GLM's strength is breadth and access to weights across task domains. Its challenges are version heterogeneity, the true cost of large checkpoints, harness sensitivity and non-identical benchmark conditions. Oblivion should build a model-independent router rather than couple its product to one model name.

Scope: publicly traceable Z.ai models and products as of 26 October 2026. APIs, released weights, historical models and adjacent products are separated.

From GLM-130B to a multimodal family

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Lineage based on the public GLM family history; specialist branches are separated in later sections.

GLM originally stands for **General Language Model**. The research family became visible in 2022 with GLM-130B. ChatGLM expanded conversational variants in 2023, while GLM-4 in 2024 moved further into dialogue, vision and tool use. Today Z.ai offers base models, APIs and complete agent products—not one homogeneous LLM line.

Period	Representative releases	Correct interpretation
2022	GLM-130B, GLM-10B	Early research lineage; not a direct production comparison with today's flagships.
2023	ChatGLM, ChatGLM2, ChatGLM3	Conversational 6B branch; mainly historical context.
2024	GLM-4, GLM-4-9B, GLM-4V-9B, GLM-4-Voice	Expansion into open weights, vision and speech.
2025	GLM-4.5/Air, 4.6, 4.7, V series	Large MoE, hybrid reasoning and agentic coding.
2026	GLM-5, 5.1, 5.2, 5.3, Flash	Long-horizon work, million-token context and new multimodality.

> **Scope of "all GLM models"**
>
> Z.ai's public hub contains many BF16/FP8 artifacts, base models and specialist checkpoints. This report covers identifiable official families; it does not count every quantization, third-party fine-tune or provider alias as a new generation.

GLM-4-Voice, GLM-Realtime, AutoGLM, GLM-PC and AutoGLM-Phone are adjacent product or agent systems, not all text LLMs. CogVideoX belongs to Z.ai's wider portfolio but is not automatically a GLM model.

The GLM-4.5 to GLM-5.3 lineage

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GLM-5.3

744B / 40B active

TEXT · LONG-HORIZON

Flagship

GLM-5.3-FLASH

320B / 18B active

MULTIMODAL · MoE

New base

GLM-4.7-FLASH

31B / 3B active

LOCAL TESTING

Compact family

Sizes and active-parameter counts are drawn from official model cards and repositories; active parameters are not total weight storage.

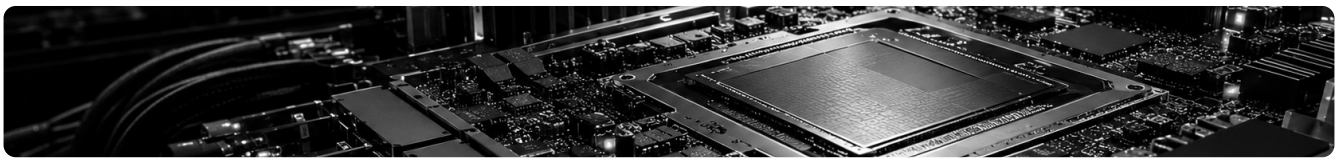
Model	Headline specification	Engineering role
GLM-4.5	355B total / 32B active; MoE	Agentic, reasoning and coding foundation; thinking and non-thinking modes.
GLM-4.5-Air	106B total / 12B active	Smaller sibling, but still not a laptop-class model.
GLM-4.6	About 355B-A32B; up to 200K	Improved coding, search, tool use and agent workflows.
GLM-4.7	About 355B-A32B	Further gains on selected coding and tool benchmarks.
GLM-4.7-Flash	About 30B-A3B	Smaller text branch; more plausible for quantized local tests.
GLM-5	About 744B / 40B active; 28.5T training tokens	Complex systems engineering and long-horizon tasks.
GLM-5.1	Agentic-coding flagship; documented 200K	Multi-step coding; long autonomous runs depend on the harness.
GLM-5.2	One-million-token context	Long-horizon work and long-context efficiency.
GLM-5.3	About 744–753B / 40B active; 1M	Same base as 5.2, with stronger post-training.
GLM-5.3-Flash	320B total / 18B active; 1M	New base and multimodal; Flash does not mean 18B total.

> **Why GLM-4.7-Flash still matters**
>
> Its total checkpoint is much smaller than GLM-5.3-Flash. It may be the more practical starting point for constrained agentic-coding experiments, but quality, memory and speed must be measured with a pinned quantization and runtime.

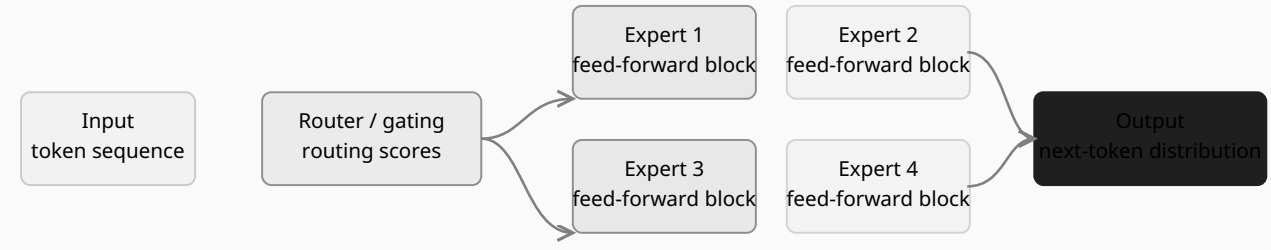
Record model ID, checkpoint date, BF16/FP8 format, context, license and endpoint together. “Latest” aliases can change over time or across providers.

MoE, sparse attention and long context

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Conceptual computing hardware visual; the multimodal diagram classifies capabilities rather than exposing internal model routing.



Conceptual schematic redrawn from the GLM-5 technical report; not a literal layer-by-layer topology.

GLM-4.5 and GLM-5 flagships use **Mixture-of-Experts**. A router activates only a subset of experts per token, reducing token-level computation versus a dense model of similar total size. Inactive experts remain in the checkpoint, however; the full weights must still be available in memory or through offload.

Property	GLM-5.3	GLM-5.3-Flash
Total scale	About 744B in README; 753B in HF listing	About 320B in NVIDIA / HF listings
Active parameters	About 40B	About 18B
Context	1,048,576 tokens in model card	1M advertised; full-length quality and throughput require tests
Input	Text	Text and images; video depends on endpoint
Key difference	5.2 base with stronger post-training	New base, 30T multimodal tokens, sparse + linear attention and mHC

Generation-5 mechanisms

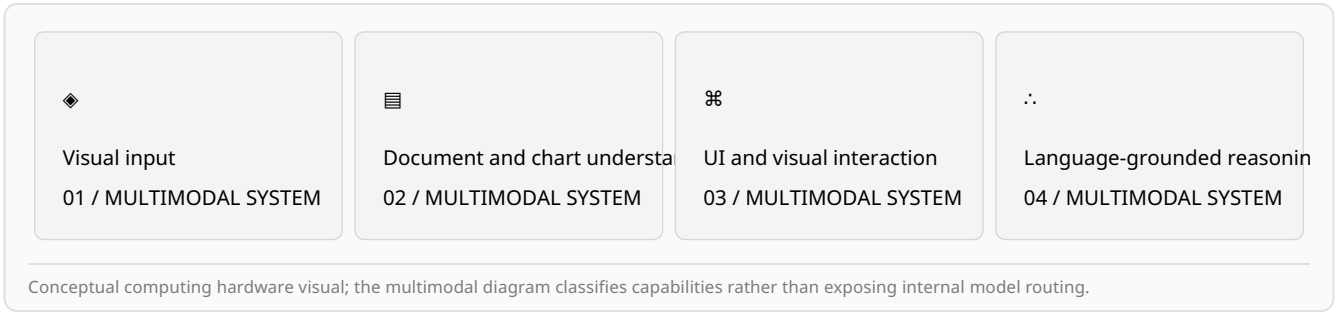
- **DSA:** Selects relevant portions of long context instead of dense attention over every token.
- **IndexShare:** Shares an indexer across selected GLM-5.2 layers; the 2.9× FLOPs reduction is a vendor-reported result under specific conditions.
- **Hybrid attention:** GLM-5.3-Flash combines sparse and linear paths to lower long-context serving cost.
- **mHC and MTP:** Hyper-connection design and multi-token prediction support scaling and speculative decoding.

> **Do not equate capacity with performance**
>
> A one-million-token ceiling does not prove cheap, fast or accurate execution at the full length. Measure sustained throughput, distant retrieval, KV cache, batch size and memory separately. Some serving implementations validate only shorter contexts.

The public GLM-5.3 listings show 744B and 753B totals under different artifact/model-card conventions. This report preserves both published values rather than silently choosing one.

GLM-V and GLM-5V-Turbo

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Conceptual computing hardware visual; the multimodal diagram classifies capabilities rather than exposing internal model routing.

Model	Scale / generation	Target capability
GLM-4.1V-9B-Thinking	About 9B; 64K	Visual reasoning over images, charts and documents.
GLM-4.5V	About 108B	Image, video and document understanding, visual grounding and GUI.
GLM-4.6V	About 108B; Flash about 9–10B	Interleaved image-text, long documents, charts and UI recreation.
GLM-5V-Turbo	API multimodal-coding foundation	Text, image, video and file input; visual coding and GUI workflows.
GLM-5.3-Flash	320B total / 18B active	First natively multimodal GLM-5 model per Z.ai; text and image input.

Where vision models are useful

- Screenshot, chart and interface-layout understanding;
- Extraction from scanned pages and documents;
- Turning visual evidence into interface code;
- Frame or video analysis where the endpoint explicitly supports it.

> **Oblivion implementation note**
>
> Preserve the original image, dimensions, crops, preliminary OCR and source file. Verify model-proposed clicks or file changes with deterministic tools before execution.

GLM-4.1V and Flash variants are more plausible for local experiments than 108B VLMs. Persian text in images, RTL documents, small print and dense tables still need Oblivion-specific tests.

OCR, speech, TTS and image generation

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OCR
Document structure

ASR / TTS
Speech input and output

IMAGE
Visual synthesis

AGENT
Computer interaction

Model	Domain	Technical note
GLM-OCR	OCR and document understanding	About 0.9B, MIT; vision encoder and lightweight decoder. 94.62 OmniDocBench is a model-card figure.
GLM-ASR-Nano-2512	Speech-to-text	About 1.5B; compact ASR. Validate on your actual languages and microphones.
GLM-TTS	Text-to-speech	Speech-token plus Flow-Matching pipeline; voice cloning requires consent and privacy controls.
GLM-Image	Image generation/editing	Hybrid autoregressive and diffusion architecture; not a conversational LLM.
AutoGLM / AutoGLM-Phone	GUI and mobile agents	Interface observation and actions; require scope limits, human approval and audit.
GLM-4-Voice / GLM-Realtime	Speech and real-time interaction	Adjacent historical/API branches; verify actual endpoint capabilities.

Cost-aware choice

For PDF extraction, GLM-OCR or a specialist OCR pipeline is generally more sensible than sending a full document to a 700B flagship. Preserve page references, coordinates and confidence with extracted text.

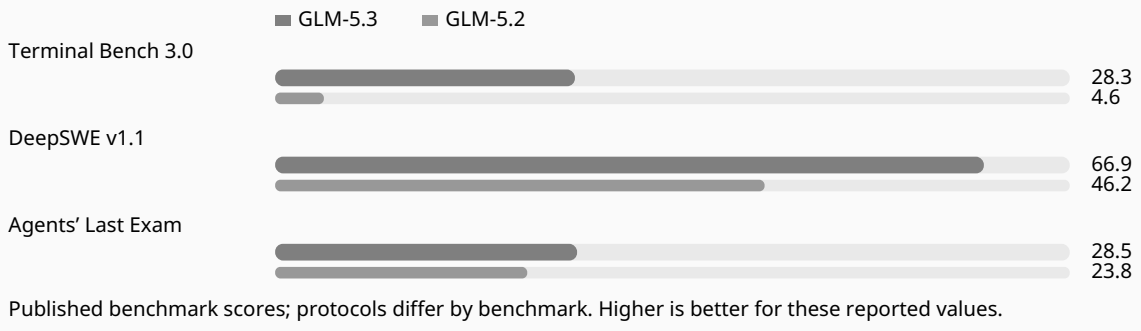
Voice privacy

Use only voices with explicit, documented permission for synthesis or cloning. Raw audio and voice embeddings are sensitive data and require retention and access controls.

CogVideoX and other Z.ai video-generation projects belong to the wider portfolio; do not label them GLM models without an explicit source.

Vendor claims versus independent assessment

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Published benchmark scores; protocols differ by benchmark. Higher is better for these reported values.

Redrawn by Oblivion from Z.ai-published figures; not an independent benchmark run.

Z.ai reports selected gains for GLM-4.7 on SWE-bench and Terminal-Bench. GLM-5.1 emphasizes agentic coding, 5.2 one-million-token context, and 5.3 stronger post-training on the 5.2 base. Z.ai reports a 50% gain over 5.2 on its internal GLM-5.3 Code Bench; that is a vendor result, not an independent replication.

Source	Reported result	Interpretation limit
Z.ai · GLM-4.7	Improvements on SWE-bench and Terminal-Bench	Vendor evaluation; harness and settings matter.
Z.ai · GLM-5.1	SWE-Bench Pro, NL2Repo and Terminal-Bench 2.0	Long workflows; multi-hour autonomy is not a product guarantee.
Z.ai · GLM-5.2	One-million-token context	Context length is not retrieval accuracy or low latency.
Z.ai · GLM-5.3	50% improvement on internal Code Bench vs 5.2	Internal company result, not independent replication.
NIST / CAISI · 17 Sep 2026	Strongest open-weight model in its cyber suite; about four months behind U.S. frontier on its aggregate index	Independent but cyber-scoped; not a full capability score.

> **Scientific standard**
>
> Each score needs benchmark/version, harness, permitted tools, reasoning budget, run count, success criteria and uncertainty. Averaging incompatible provider leaderboards is not a defensible scientific ranking.

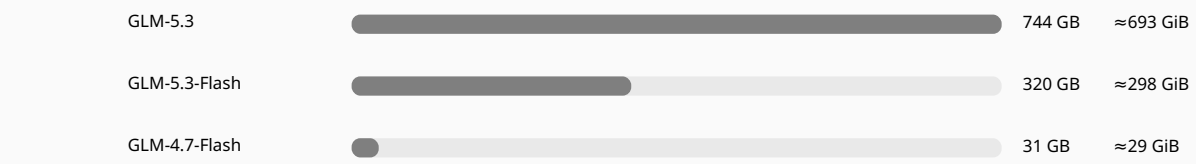
CAISI evaluated the model as an agent in a specified ReAct harness. Its finding applies to that setup, not as an invariant property of every deployment.

Open weights do not mean easy local execution

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Conceptual infrastructure image for the report narrative; not verified photography of a Z.ai data center.



Simple estimate from total parameters at one byte per parameter; excludes KV cache, runtime, sharding overhead and activations.

GLM-5 flagships have hundreds of billions of parameters. BF16 weights require hundreds of gigabytes to more than a terabyte of raw memory. FP8 reduces the footprint but still leaves a very large model. Active parameters explain part of token compute, not the memory required for every weight.

Model	Local deployment reality	Practical recommendation
GLM-5.3 / 5.2 / 5.1	Multi-GPU infrastructure and engineered serving	Server or research cluster, not a typical laptop.
GLM-5.3-Flash	About 320B total despite 18B active	Not an 18B local model; reported FP8 weights are around 300 GiB before KV/runtime.
GLM-4.5-Air / 4.7	Large datacenter-class checkpoints	Quantization/offload may help; measure memory and latency.
GLM-4.7-Flash	30B total / 3B active	More plausible for quantized local tests.
GLM-4.1V-9B / GLM-OCR	Relatively smaller	Better laptop candidates, subject to actual VRAM and runtime.

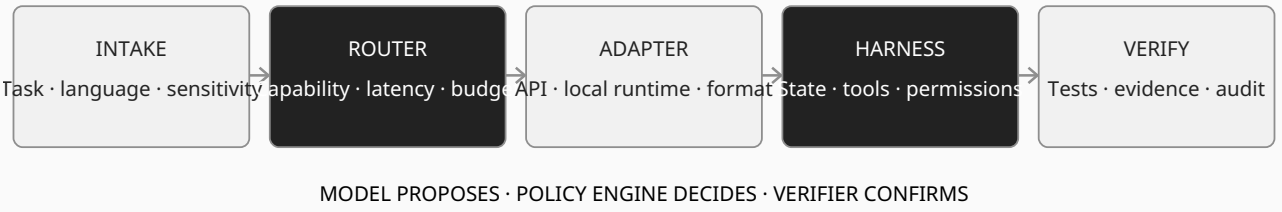
Measure before deployment

- Exact weight-file size and peak VRAM/RAM;
- Prefill/decode tokens/sec and time to first token;
- p50/p95 latency at documented batch and context;
- Persian quality, tool error rate and cost per task;
- Stability at long context and concurrent load.

> **Oblivion hardware guidance**
>
> On an RTX 3060-class laptop, full-weight GLM-5.3 or 5.3-Flash is not a realistic local target. Test GLM-OCR, GLM-4.1V-9B and smaller quantized models locally; serve flagships through an endpoint or a multi-GPU host. Record actual VRAM and checkpoint before claiming compatibility.

A model-independent router and three critique rounds

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A model-independent architecture for Oblivion

Workload	Candidate to evaluate	Reason
Difficult coding and long planning	GLM-5.3	High-end flagship; large infrastructure.
Lower-cost agent work	GLM-5.3-Flash or 5.1	Compare in your harness; Flash still has huge total weights.
Local text	GLM-4.7-Flash	Smaller total checkpoint; test quantization.
Vision and GUI	GLM-4.1V-9B or 4.6V Flash	Dedicated visual route.
PDF and scans	GLM-OCR	Specialist instead of text flagship.
Speech and images	GLM-ASR / TTS / Image	Separate modules and data policies.

Three rounds of critique

Round 1: Newest does not mean best; route by task. **Round 2:** 18B active does not mean 18B total; GLM-5.3-Flash is about 320B total. **Round 3:** Agentic capability is not a security guarantee; enforce policy outside the model and gate side effects.

> **Final Oblivion recommendation**
>
> Evaluate GLM-5.3 as a high-end reasoning/coding engine; treat 5.3-Flash as large multimodal infrastructure rather than a laptop model; use 4.7-Flash for local tests and OCR/ASR/TTS/V models for specialist tasks. The model proposes actions; an independent gateway grants or denies permission.

Pre-release protocol

1. Build 100–200 anonymized, multilingual real tasks;
2. Pin checkpoint, quantization, runtime, prompt and budget;
3. Measure pass rate, tool errors, hallucination, injection resistance and latency;
4. Repeat unstable tasks at least three times;
5. Select a Pareto balance across quality, cost, speed and risk.

Scope, limitations and traceability

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RESEARCH TRACEABILITY PRIMARY SOURCES · MODEL CARDS · TECHNICAL PAPERS

This report uses Z.ai research pages, GitHub repositories, Z.ai Hugging Face model cards, vendor technical material and the independent CAISI/NIST assessment. Vendor benchmark claims are labeled as Z.ai-reported; NIST findings are an independent assessment with a cyber-specific scope. Benchmarks were not independently rerun on Oblivion hardware. File size, endpoint support, practical context and licensing can vary by checkpoint, provider and runtime.

Primary sources

1. Z.ai Research Hub https://www.zhipuai.cn/en/research
2. Z.ai GLM-5.3 announcement https://z.ai/blog/glm-5.3
3. Official GLM-5 repository https://github.com/zai-org/GLM-5
4. Z.ai model and checkpoint index https://huggingface.co/zai-org/models
5. NVIDIA GLM-5.3-Flash model card https://build.nvidia.com/z-ai/glm-5-3-flash/modelcard
6. NIST / CAISI cyber assessment, 17 Sep 2026 https://www.nist.gov/news-events/news/2026/09/caisis-assessment-zais-glm-53-cyber-capabilities
7. GLM-V repository https://github.com/zai-org/GLM-V
8. GLM-OCR repository https://github.com/zai-org/GLM-OCR
9. GLM-ASR-Nano-2512 https://huggingface.co/zai-org/GLM-ASR-Nano-2512
10. GLM-TTS repository https://github.com/zai-org/GLM-TTS
11. GLM-Image repository https://github.com/zai-org/GLM-Image
12. GLM-4.5 technical report https://arxiv.org/abs/2508.06471
13. GLM-4.1V-Thinking paper https://arxiv.org/abs/2507.01006

Boundary of the conclusion

This is an independent synthesis of public material, not a security certification, reproduced benchmark, Z.ai endorsement or guarantee of suitability for a production Oblivion system.

Prepared for the Oblivion engineering team · Version 1.0